3.2.2. Magnetic field at a point along the equatorial line due to a magnetic dipole (bar magnet)

Consider a bar magnet NS as shown in Figure 3.14. Let N be the north pole and S be the south pole of the bar magnet, each with pole strength q_m and separated by a distance of $2l$. The magnetic field at a point C (lies along the equatorial line) at a distance r from the geometrical centre O of the bar magnet can be computed by keeping unit north pole (q_{mC} = 1 A m) at C.

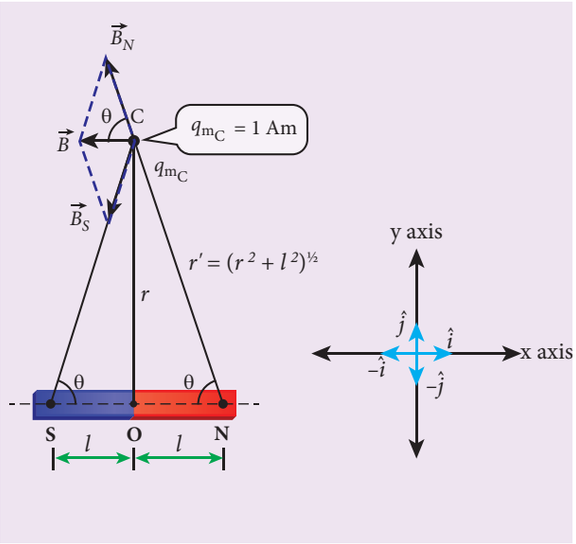


Figure 3.14 Magnetic field at a point along the equatorial line due to a magnetic dipole

The magnetic field at C due to the north pole is

$$\vec{B}_N = -B_N \cos\theta \hat{i} + B_N \sin\theta \hat{j} \qquad (3.15)$$

where $B_N = \dfrac{\mu_\circ}{4\pi} \dfrac{q_m}{r'^2}$

Here $r' = (r^2 + l^2)^{\frac{1}{2}}$

The magnetic field at C due to the south pole is

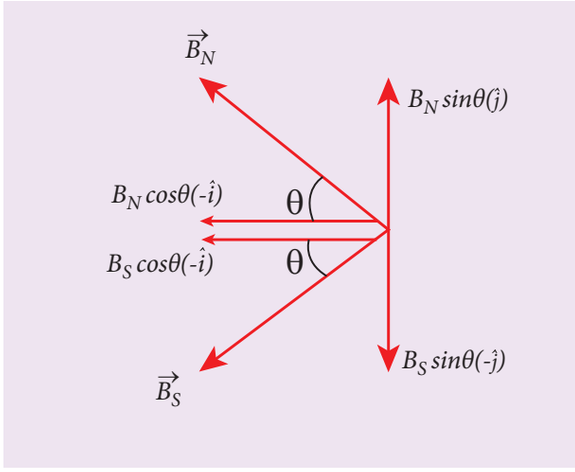


Figure 3.15 Components of magnetic field

$$\vec{B}_S = -B_S \cos\theta \hat{i} - B_S \sin\theta \hat{j} \qquad (3.16)$$

where, $B_S = \dfrac{\mu_\circ}{4\pi} \dfrac{q_m}{r'^2}$

From equations (3.15) and (3.16), the net magnetic field at point C due to the dipole is $\vec{B} = \vec{B}_N + \vec{B}_S$.

$$\vec{B} = -(B_N + B_S)\cos\theta \hat{i} \quad \text{Since, } B_N = B_S$$

$$\vec{B} = -\frac{2\mu_\circ}{4\pi} \frac{q_m}{r'^2} \cos\theta \hat{i} = -\frac{2\mu_\circ}{4\pi} \frac{q_m}{(r^2 + l^2)} \cos\theta \hat{i} \qquad (3.17)$$

In a right angle triangle NOC as shown in Figure 3.14

$$\cos\theta = \frac{adjacent}{hypotenuse} = \frac{l}{r'} = \frac{l}{\left(r^2 + l^2\right)^{\frac{1}{2}}} \qquad (3.18)$$

Substituting equation (3.18) in equation (3.17), we get

$$\vec{B} = -\frac{\mu_\circ}{4\pi} \frac{q_m \times (2l)}{(r^2 + l^2)^{\frac{3}{2}}} \hat{i} \qquad (3.19)$$

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